Effects of a low-fat, high-fiber diet and exercise program on breast cancer risk factors in vivo and tumor cell growth and apoptosis in vitro.

The present study investigated the effects of a diet and exercise intervention on known breast cancer (BCa) risk factors, including estrogen, obesity, insulin, and insulin-like growth factor-I (IGF-I), in overweight/obese, postmenopausal women. In addition, using the subjects' pre- and postintervention serum in vitro, serum-stimulated growth and apoptosis of three estrogen receptor-positive BCa cell lines were studied. The women where placed on a low-fat (10-15% kcal), high-fiber (30-40 g per 1,000 kcal/day) diet and attended daily exercise classes for 2 wk. Serum estradiol was reduced in the women on hormone treatment (HT; n = 28) as well as those not on HT (n = 10). Serum insulin and IGF-I were significantly reduced in all women, whereas IGF binding protein-1 was increased significantly. In vitro growth of the BCa cell lines was reduced by 6.6% for the MCF-7 cells, 9.9% for the ZR-75-1 cells, and 18.5% for the T-47D cells. Apoptosis was increased by 20% in the ZR-75-1 cells, 23% in the MCF-7 cells, and 30% in the T-47D cells (n = 12). These results show that a very-low-fat, high-fiber diet combined with daily exercise results in major reductions in risk factors for BCa while subjects remained overweight/obese. These in vivo serum changes slowed the growth and induced apoptosis in serum-stimulated BCa cell lines in vitro.